



BITS Pilani
Dubai Campus

CAA Criteria 2 Learning Outcomes



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CAA/BPDC

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OUTLINE

CLOs: SMART Learning Outcomes

CLOs & PLOs Mapping

Evaluation Components (ECs)



CLOs/ECs Weightage Distribution(%)

QP Setting Guidelines

CLOs Attainment

PLOs Attainment



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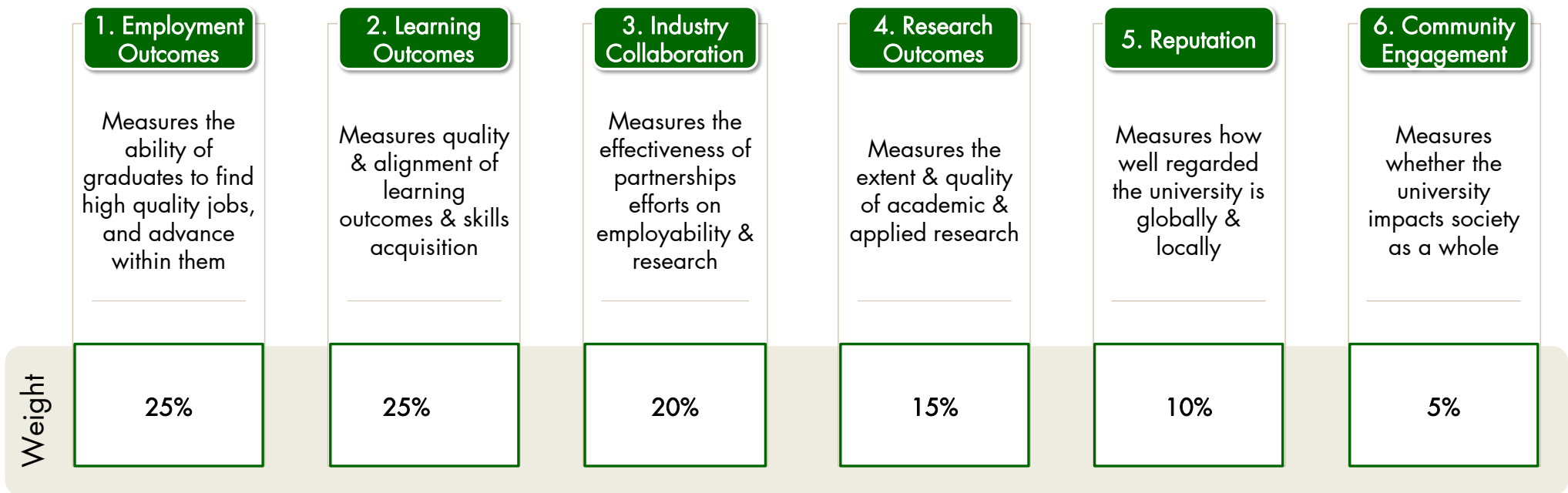
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Outcome Based Evaluation Framework



Ref: <https://caa.ae/>



Outcome Based Evaluation Framework



CLO Guidelines



✚ Number of CLOs:

- 📌 Each course should have **4 to 6 CLOs**.
- 📌 Fewer → doesn't cover enough depth.
- 📌 More → becomes unmanageable for mapping and assessment.

✚ Verb Rule:

- 📌 Use **one Bloom's verb per CLO** (Explain, Construct, Analyze, Design, Develop...).
- 📌 Don't stack verbs like "*Design and evaluate and analyze*".

✚ Undergraduate CLOs:

- 📌 **First Year (1–2):** CLO verbs mostly **RBT 1–3** (*Remember, Understand, Apply*).
- 📌 **Second/Third Year theory/technical courses (Year 2–3):** CLO verbs up to **RBT 4 (Analyze)**.
- 📌 **Final-year / project / advanced electives:** Can reach **RBT 5–6 (Evaluate, Create)**.

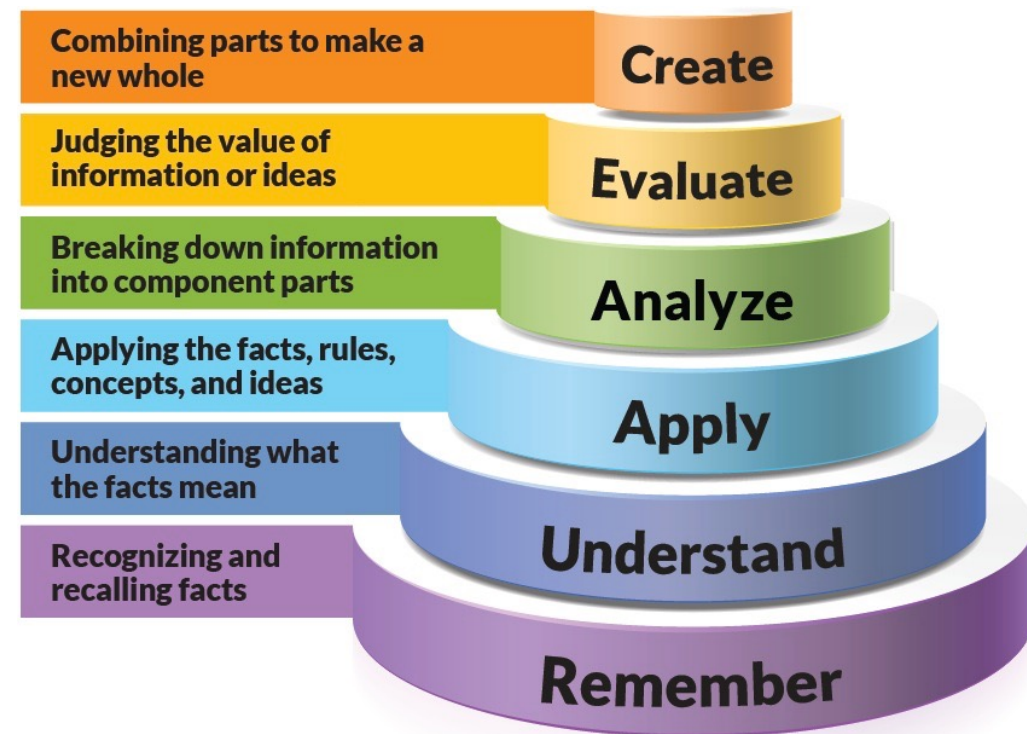
✚ Postgraduate CLOs:

- 📌 Should emphasize **higher-order levels** (Analyze, Evaluate, Create).
- 📌 Less focus on *Remember/Understand*, since PG assumes prior foundation.
- 📌 CLOs should reflect **research orientation, critical thinking, originality**.

Bloom's Taxonomy



- 📌 **Remembering:** To recall facts, basic concepts, or retrieval of material
- 📌 **Understanding:** To explain ideas, concepts, or construct meaning from written material or graphics.
- 📌 **Applying:** To use information in new situations such as models, diagrams, or presentations.
- 📌 **Analyzing:** To draw connections among ideas, concepts, or determining how each part interrelate to an overall structure or purpose
- 📌 **Evaluating:** To justify a stand or decision; to make judgements based on criteria and standards through checking and critiquing.
- 📌 **Creating:** To produce new or original work.



<https://tips.uark.edu/using-blooms-taxonomy/#gsc.tab=0>

Bloom's Taxonomy



Bloom's Level	Key Verbs (keywords)	Example Learning Objective
Create	design, formulate, build, invent, create, compose, generate, derive, modify, develop.	<i>By the end of this lesson, the student will be able to design an original homework problem dealing with the principle of conservation of energy.</i>
Evaluate	choose, support, relate, determine, defend, judge, grade, compare, contrast, argue, justify, support, convince, select, evaluate.	By the end of this lesson, the student will be able to determine whether using conservation of energy or conservation of momentum would be more appropriate for solving a dynamics problem.
Analyze	classify, break down, categorize, analyze, diagram, illustrate, criticize, simplify, associate.	<i>By the end of this lesson, the student will be able to differentiate between potential and kinetic energy.</i>
Apply	calculate, predict, apply, solve, illustrate, use, demonstrate, determine, model, perform, present.	<i>By the end of this lesson, the student will be able to calculate the kinetic energy of a projectile.</i>
Understand	describe, explain, paraphrase, restate, give original examples of, summarize, contrast, interpret, discuss.	<i>By the end of this lesson, the student will be able to describe Newton's three laws of motion in her/his own words</i>
Remember	list, recite, outline, define, name, match, quote, recall, identify, label, recognize.	<i>By the end of this lesson, the student will be able to recite Newton's three laws of motion.</i>

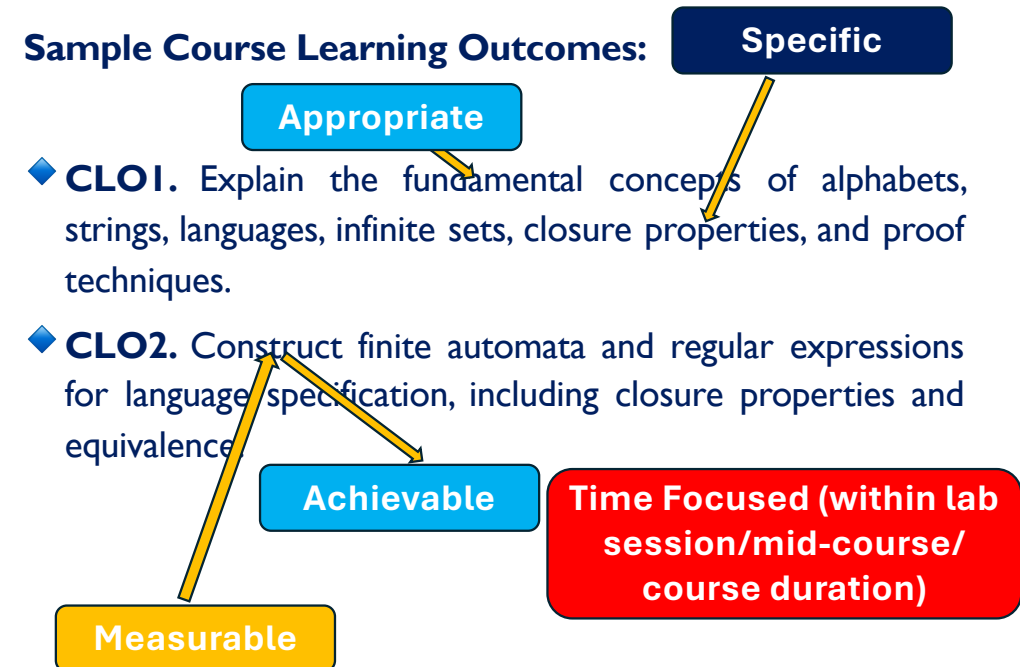
<https://tips.uark.edu/using-blooms-taxonomy/#gsc.tab=0>

SMART Learning Outcomes



- ✚ **S: Specific**
 - ◆ Clearly state what students are expected to know/do.
- ✚ **M: Measurable**
 - ◆ Use measurable action verbs (e.g., analyze, build, compare).
 - ◆ Avoid vague verbs (know, understand)
- ✚ **A: Appropriate** (Attainable, Achievable)
 - ◆ Students should realistically be able to achieve the outcomes
- ✚ **R: Realistic** (Relevant)
 - ◆ Outcome should align with program mission, goals, and course aims.
- ✚ **T: Time-focused** (Time-limited, Testable)
 - ◆ Should be achievable within given time constraints.

Sample Course Learning Outcomes:



SMART Learning Outcomes



Unmeasurable (Verbs to avoid):

Unmeasurable and Nonspecific Verbs	Measurable and Specific Verbs
Understand	Recognize, Describe
Appreciate	Debate
Be familiar	Recognize, Identify, Discuss
Become aware of	Recognize, Identify, Discuss
Consider	Appraise
Continue	Not Applicable
Know	Recognize, Describe
Review	Discuss, Examine
Learn	Not suitable for LOs
Deepen	Not suitable for LOs
Ensure	Not suitable for LOs

Explain → student writes a short note, answers “what/why” in an exam.

***Understand* → How do we prove “understanding”? It’s indirect**

Instead, replace with measurable verbs like *Explain, Describe, Summarize*

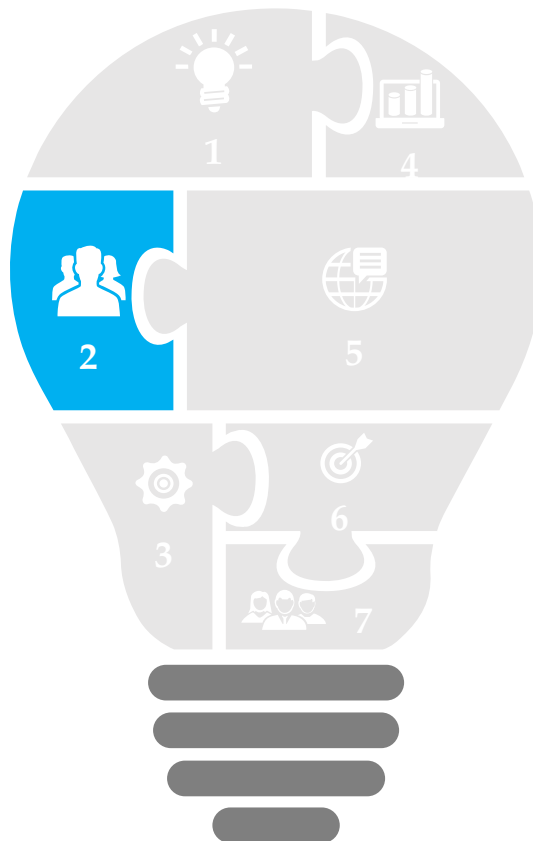


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📌 Mapping Rule:

- ◆ Each **CLO** must map to at least one **PLO**.
- ◆ **Max 2 PLOs per CLO** (only if clearly justifiable).
- ◆ Avoid mapping one CLO to 3+ PLOs — avoid “force-fitting.”

📌 Coverage Rule:

- ◆ Not all PLOs need to be covered in every course.
- ◆ Across the **whole program**, all PLOs will be addressed.

📌 CLO → PLO Mapping:

- ◆ 0/1 → if the CLO contributes to more than 60% of the specific PLO

CLO_PLO Mapping								
CLO/PLO	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8
CLO1		1						
CLO2		1						
CLO3		1						
CLO4		1		1				
CLO5		1		1				

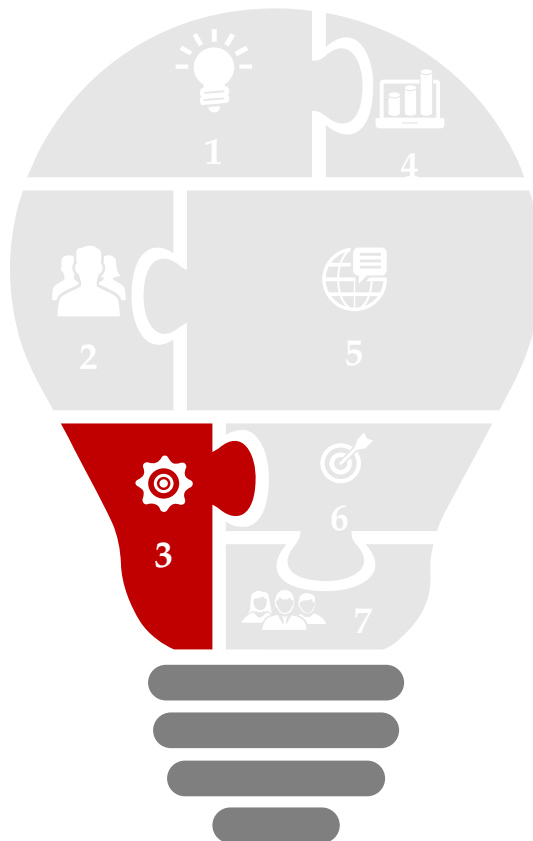


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CLOs/PLOs Assessment



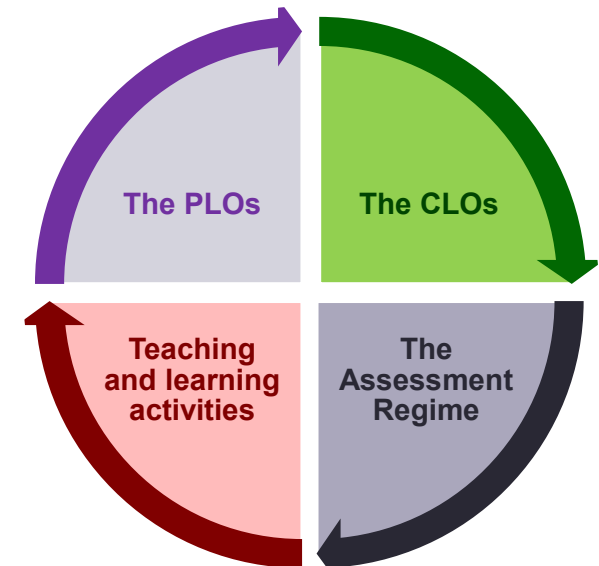
📌 To assess PLOs, you need assess CLOs. This can be done via:

◆ Direct Assessment:

- ◆ Student Assessment in courses and other learning activities (Exams and Rubrics)
 - ◆ **Formative Assessments (Min 30%):** Quiz 1, Quiz 2, Assignment, Exit exams or progress tests
 - ◆ **Summative Assessments (Max 70%):** Midterm, Comprehensive Exam

◆ Indirect Assessment:

- ◆ Learning Outcomes Surveys for perception of student attainment of learning outcomes
- ◆ Alumni and Employers Surveys





Evaluation Components

EC No	Evaluation Components	Weightage	Duration	Date & Time	Comments
1	Quiz	10%	20 minutes	30.09.2025 T2	Closed book
2	Mid Sem Test	25%	90 minutes	27.10.2025 AN	Closed book
3	Exit Quiz*	5%	10-15 minutes	N/A (Continuous)	Open Book, Online
4	Problem Solving*	5%	10-15 minutes	N/A (Continuous)	Open Book, Group Participation
5	Assignment	20%	2 days	09.12.2025 to 10.12.2025	Open Book
6	Comprehensive Exam	35%	2 hours	22.12.2025 AN	Closed book

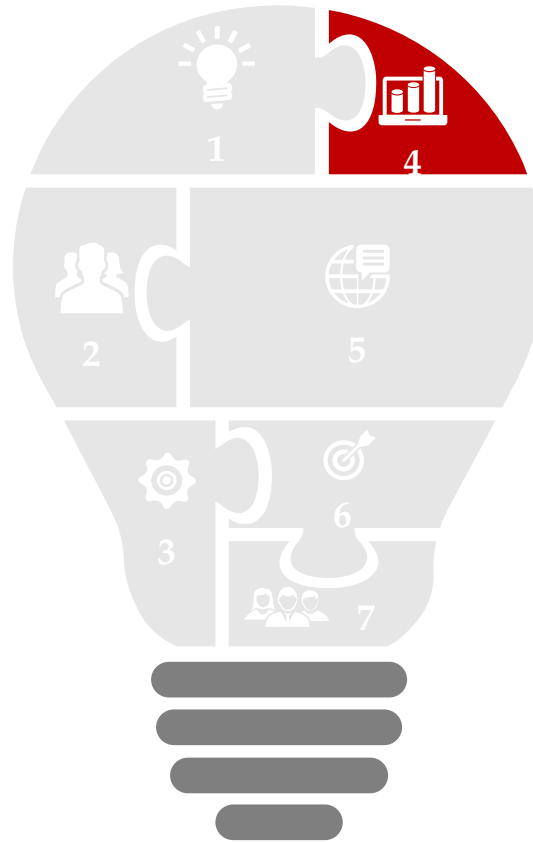


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Weightage Distribution

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5	Assignment	20%	2 days	09.12.2025 to 10.12.2025	Open Book
6	Comprehensive Exam	35%	2 hours	22.12.2025 AN	Closed book

CLOS/Evaluation Component	Weightage Distribution (%)						Total
	EC1 (10%)	EC2 (25%)	EC3 (5%)	EC4 (5%)	EC5 (20%)	EC6 (35%)	
CLO1	60	30				10	100
CLO2		30	40		20	10	100
CLO3		40		50		10	100
CLO4				50	20	30	100
CLO5					60	40	100

CLOS/Evaluation Component	Mark Distribution (200)						Total
	EC1 (20)	EC2 (50)	EC3 (10)	EC4 (10)	EC5 (40)	EC6 (70)	
CLO1	20	15				7	42
CLO2		15	10		8	7	40
CLO3		20		5		7	32
CLO4				5	8	21	34
CLO5					24	28	52
Total	20	50	10	10	40	70	200

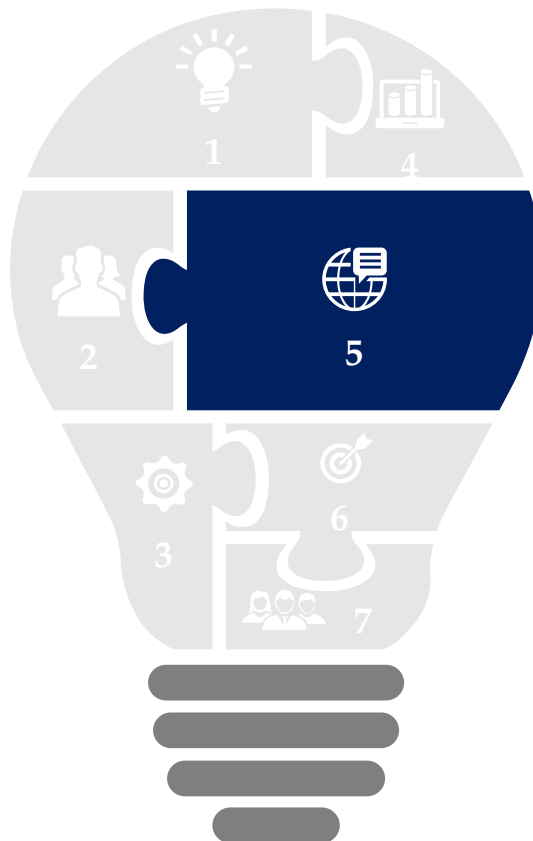


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Question Paper Guidelines



📌 **CLO-RBT Alignment:** Every question should be mapped with a **CLO** and its corresponding **Bloom's Taxonomy Level (RBT Level)**.

◆ **CLO Mapping:**

- ◆ In the QP, mention the CLO number next to each question
- ◆ Ensure the marks evenly distributed across the CLOs as mentioned in the weightage Distribution.

◆ **RBT Mapping:**

- ◆ In the QP, mention the RBT Level number next to each question.
- ◆ Ensure **diversity of cognitive levels** in questions by incorporating:
 - ◆ **Lower-order thinking skills (LOTS):** Remembering (RBT 1), Understanding (RBT 2), and Applying (RBT 3).
 - ◆ **Higher-order thinking skills (HOTS):** Analyzing (RBT 4), Evaluating (RBT 5), and Creating (RBT 6).

📌 [Sample Question Paper](#)

Evaluation Process & Rubrics



- 📌 **Clarity:** Each assessment must have well-defined criteria for scoring.
- 📌 **Standardization:** Uniformity in grading across faculty members.
- 📌 **Differentiation:**
 - ◆ **Formative Assessments (Continuous Evaluation)** – Quiz 1, Quiz 2, Assignment
 - ◆ **Summative Assessments (Final Evaluation)** – Midterm, Comprehensive Exam
- 📌 **Design Rubrics for Each Component**



Sample Question Formats

Question Types	Questions	CLO	RBT Level	Marks
Q1 (MCQ)	Which of the following sets is not a language? a) $\{0,1\}^*$ b) $\{\epsilon\}$ c) $\{a^n b^n \mid n \geq 0\}$ d) $\{\text{red, blue}\}$	CLO1	Remember (L1)	2
Q2 (Fill-in)	The concatenation of $L1 = \{a, b\}$ and $L2 = \{ab\}$ is _____.	CLO1	Apply (L3)	2
Q3 (True/False + Justify)	The union of two regular languages is always regular. (True / False? Justify)	CLO1	Understand (L2)	3
Q4 (Short Answer)	Define an infinite set with one example and explain why it is infinite.	CLO1	Understand (L2)	3
Q5 (Short Problem)	Write a regular expression for the language of all binary strings containing exactly one "1".	CLO1	Apply (L3)	4
Q6 (Mini Problem)	Construct a DFA over $\{0,1\}$ that accepts all strings ending with "10". Draw the transition diagram.	CLO2	Apply (L3)	3
Q7 (Analytical Problem)	Minimize the given DFA (diagram provided) and state the minimal number of states.	CLO2	Analyze (L4)	3



Sample Rubric

Category	Status	LEX Program	YACC Program	Total Marks
I	Complete Implementation and Full Output (check logic)	10	20	30
II	Full Implementation but Partial Output (check logic)	10	15	25
III	Full Implementation and No Output (check logic)	8	12	20
IV	Lex alone compiles with proper logic & syntax. YACC not done	8	0	8
V	YACC alone compiles with proper logic & syntax. Lex not done	0	8	8
VI	Compilation Errors and No Output (check logic)	2,3,4	2, 3, 4	4, 6, 8
VII	Student not trying the problem / Sitting Idle. No updates done to lex.l and yacc.y	0	0	0

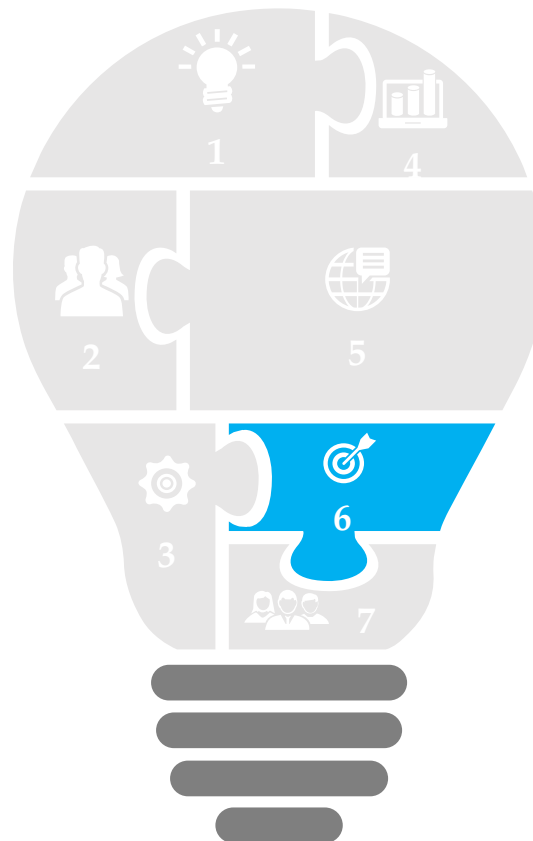


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Mark Entry

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE PILANI, DUBAI CAMPUS

DUBAI INTERNATIONAL ACADEMIC CITY, DUBAI

MARKS & GRADE CARD

XX SEMESTER 20XX - 20XX

COURSE: XXXX

COURSE NO.: XXXX

INSTRUCTOR: XXXX

INSTRUCTOR IN-CHARGE: XXXX

S.NO.	ID NO	NAME	CLO Total					Quiz - I (20 Marks)
		Marks	14	6	0	0	0	
		CLO	CLO1	CLO2	CLO3	CLO4	CLO5	
1			8.5	4.5	0	0	0	13.00
2			5.5	2.5	0	0	0	8.00
3			8.5	4.5	0	0	0	13.00
4			11	5	0	0	0	16.00
5			8.5	4.5	0	0	0	13.00
6			8.5	4.5	0	0	0	13.00
7			5.5	2.5	0	0	0	8.00
8			12	6	0	0	0	18.00
9			5.5	2.5	0	0	0	8.00
10			6.5	3.5	0	0	0	10.00
11			8.5	4.5	0	0	0	13.00
12			12	6	0	0	0	18.00
13			8.5	4.5	0	0	0	13.00
14			11	5	0	0	0	16.00
15			8.5	4.5	0	0	0	13.00
16			4.5	3	0	0	0	7.50
17			8.5	4.5	0	0	0	13.00
18			12	6	0	0	0	18.00
19			5.5	2.5	0	0	0	8.00
20			6.5	3.5	0	0	0	10.00





CLO-Level Alignment

S.NO.	ID NO	NAME	CLO Total					Quiz - I (20 Marks)
		Marks	14	6	0	0	0	
		CLO	CLO1	CLO2	CLO3	CLO4	CLO5	
1			8.5	4.5	0	0	0	13.00
2			5.5	2.5	0	0	0	8.00
3			8.5	4.5	0	0	0	13.00
4			11	5	0	0	0	16.00
5			8.5	4.5	0	0	0	13.00
6			8.5	4.5	0	0	0	13.00
7			5.5	2.5	0	0	0	8.00
8			12	6	0	0	0	18.00
9			5.5	2.5	0	0	0	8.00
10			6.5	3.5	0	0	0	10.00
11			8.5	4.5	0	0	0	13.00
12			12	6	0	0	0	18.00
13			8.5	4.5	0	0	0	13.00
14			11	5	0	0	0	16.00
15			8.5	4.5	0	0	0	13.00
16			4.5	3	0	0	0	7.50
17			8.5	4.5	0	0	0	13.00
18			12	6	0	0	0	18.00
19			5.5	2.5	0	0	0	8.00
20			6.5	3.5	0	0	0	10.00
		CLASS HIGHEST	12	6	0	0	0	18.00
		CLASS LOWEST	4.5	2.5	0	0	0	8.00
		CLASS AVERAGE	8.00	4.00	0.00	0.00	0.00	12.00
		CLO AVERAGE	0.58	0.67	0	0	0	

Average of CLOs across ECs
 Quiz1: CLO1 = 8/14 = 0.58
 Quiz1: CLO2 = 4/6 = 0.67

CLOS/Evaluation Component	Weightage Distribution (%)				
	EC1 (10%)	EC2 (30%)	EC3 (10%)	EC4 (10%)	EC5 (40%)
CLO1	60	20		10	10
CLO2	40	40		15	5
CLO3		40	20	20	15
CLO4			55	20	25
CLO5			20	35	45

CLOS/Evaluation Component	Mark Distribution (%)					Total
	EC1 (20)	EC2 (60)	EC3 (20)	EC4 (20)	EC5 (80)	
CLO1	14	12		2	8	36
CLO2	6	24		3	4	37
CLO3		24	5	4	12	45
CLO4			11	4	20	35
CLO5			4	7	36	47
Total	20	60	20	20	80	200



CLO Attainment Calculation

CLO Attainment					
	CLO1	CLO2	CLO3	CLO4	CLO5
Quiz 1	0.58	0.67	0	0	0
Midterm	0.65	0.59	0.65	0	0
Quiz 2	0	0	0.68	0.64	0.35
Assignment	1	0.94	0.8	0.65	0.69
Compre	0.7	0.48	0.79	0.62	0.71
Attainment (Weighted Average)	0.65	0.62	0.70	0.63	0.67

Attainment Weighted Average (CLO wise)

$$= (\text{Quiz 1\%} * \text{Quiz 1 Att.}) + (\text{Midterm \%} * \text{Midterm. Att.}) + (\text{Assignment\%} * \text{Assignment Att.}) + (\text{Compre\%} * \text{Compre Att.})$$

$$= (60\% * 0.58) + (20\% * 0.65) + (10\% * 1) + (10\% * 0.7)$$

$$= 0.648$$

CLOS/Evaluation Component	Weightage Distribution (%)					Total
	EC1 (20)	EC2 (60)	EC3 (20)	EC4 (20)	EC5 (80)	
CLO1	14	12		2	8	36
CLO2	6	24		3	4	37
CLO3		24	5	4	12	45
CLO4			11	4	20	35
CLO5			4	7	36	47
Total	20	60	20	20	80	200

CLOS/Evaluation Component	Weightage Distribution (%)				
	EC1 (10%)	EC2 (30%)	EC3 (10%)	EC4 (10%)	EC5 (40%)
CLO1	60	20		10	10
CLO2	40	40		15	5
CLO3		40	25	20	15
CLO4			55	20	25
CLO5			20	35	45

Coursewise Attainment Weighted Average

$$= [(\text{CLO 1 Marks} * \text{CLO1 Att.}) + (\text{CLO 2 Marks} * \text{CLO2 Att.}) + (\text{CLO 3 Marks} * \text{CLO3 Att.}) + (\text{CLO 4 Marks} * \text{CLO4 Att.}) + (\text{CLO 5 Marks} * \text{CLO5 Att.})] / \text{Total Marks}$$

$$= [(36 * 0.65) + (37 * 0.62) + (45 * 0.70) + (35 * 0.63) + (47 * 0.67)] / 200$$

$$= 0.66$$

Course Attainment



CLO Attainment

CLO Attainment					
	CLO1	CLO2	CLO3	CLO4	CL05
Quiz 1	0.58	0.67	0	0	0
Midterm	0.65	0.59	0.65	0	0
Quiz 2	0	0	0.68	0.64	0.35
Assignment	1	0.94	0.8	0.65	0.69
Compre	0.7	0.48	0.79	0.62	0.71
Attainment (Weighted Average)	0.65	0.62	0.70	0.63	0.67

CLO Attainment					
	CLO1	CLO2	CLO3	CLO4	CL06
Quiz 1					
Midterm					
Quiz 2					
Assignment					
Compre					
Attainment (Weighted Average)	0.77	0.77	0.77	0.62	0.46

SEM I 2025-26							
Course Code	Course	CLO1	CLO2	CLO3	CLO4	CLO5	CLO6
CS 001	DAA	0.65	0.62	0.70	0.63	0.67	
CS 002	OS	0.77	0.77	0.77	0.62	0.46	0.46
CS 003	TOC	1	0.94	1	0.72		0.87
CS 004	CN	1	1	1	1	0.89	0.68

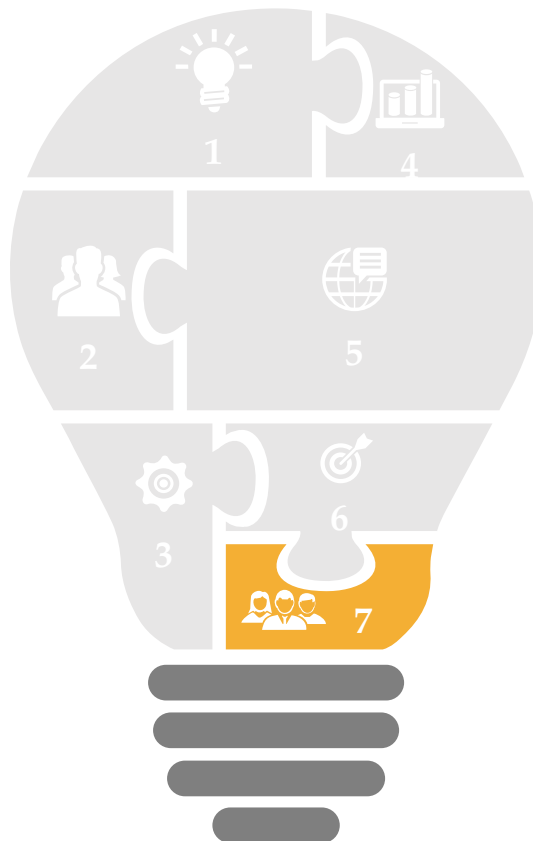


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PLO Attainment Calculation

CLO Attainment					
	CLO1	CLO2	CLO3	CLO4	CLO5
Quiz 1	0.58	0.67	0	0	0
Midterm	0.65	0.59	0.65	0	0
Quiz 2	0	0	0.68	0.64	0.35
Assignment	1	0.94	0.8	0.65	0.69
Compre	0.7	0.48	0.79	0.62	0.71
Attainment (Weighted Average)	0.65	0.62	0.70	0.63	0.67

CLO_PLO Mapping									
	CLO Attainment	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8
CLO1	0.65		1		1				
CLO2	0.62		1	1	1				
CLO3	0.70		1		1				1
CLO4	0.63		1		1		1		
CLO5	0.67		1		1				
Attainment (Sum)		0	5	1	5	0	1	0	1

$$\begin{aligned} \text{CLO_PLO} &= \text{CLO1 Attainment} * \text{PLO2} \\ &= 0.65 * 1 \\ &= 0.65 \end{aligned}$$

CLO_PLO Attainment									
	CLO Attainment	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8
CLO1	0.65	0.00	0.65	0.00	0.65	0.00	0.00	0.00	0.00
CLO2	0.62	0.00	0.62	0.62	0.62	0.00	0.00	0.00	0.00
CLO3	0.70	0.00	0.70	0.00	0.70	0.00	0.00	0.00	0.70
CLO4	0.63	0.00	0.63	0.00	0.63	0.00	0.63	0.00	0.00
CLO5	0.67	0.00	0.67	0.00	0.67	0.00	0.00	0.00	0.00
Attainment (Average)			0.65	0.62	0.65		0.63		0.70

$$\begin{aligned} \text{Attainment (Average) PLO2} &= (\text{CLO1} + \text{CLO2} + \text{CLO3} + \text{CLO4} + \text{CLO5}) / \text{PLO2} \\ &= (0.65 + 0.62 + 0.70 + 0.63 + 0.67) / 5 \\ &= 3.27 / 5 \\ &= 0.65 \end{aligned}$$



PLO Attainment

COURSE 1									
	CLO Attainment	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8
CLO1	0.65	0.00	0.65	0.00	0.65	0.00	0.00	0.00	0.00
CLO2	0.62	0.00	0.62	0.62	0.62	0.00	0.00	0.00	0.00
CLO3	0.70	0.00	0.70	0.00	0.70	0.00	0.00	0.00	0.70
CLO4	0.63	0.00	0.63	0.00	0.63	0.00	0.63	0.00	0.00
CLO5	0.67	0.00	0.67	0.00	0.67	0.00	0.00	0.00	0.00
Attainment (Average)			0.65	0.62	0.65		0.63		0.70

COURSE 2									
	CLO Attainment	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8
CLO1	0.50	0.00	0.00	0.50	0.00	0.00	0.00	0.00	0.00
CLO2	0.53	0.00	0.00	0.53	0.00	0.00	0.00	0.00	0.00
CLO3	0.60	0.00	0.00	0.00	0.60	0.00	0.00	0.00	0.60
CLO4	0.70	0.00	0.00	0.00	0.70	0.00	0.70	0.00	0.00
CLO5	0.45	0.00	0.00	0.00	0.45	0.00	0.45	0.00	0.00
Attainment (Average)				0.52	0.58		0.58		0.60

COURSE 3									
	CLO Attainment	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8
CLO1	0.80	0.80	0.00	0.00	0.00	0.00	0.00	0.00	0.00
CLO2	0.85	0.85	0.00	0.85	0.00	0.00	0.00	0.00	0.00
CLO3	0.76	0.76	0.00	2.28	0.00	0.00	0.00	0.00	0.76
CLO4	0.87	0.87	0.00	0.00	0.00	0.00	0.87	0.00	0.00
CLO5	0.79	0.79	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Attainment (Average)		0.81		0.78			0.87		0.76

SEM I 2025-26									
Course Code	Course	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8
CS 001	DAA		0.65	0.62	0.65		0.63		0.70
CS 002	OS			0.52	0.58		0.58		0.60
CS 003	TOC	0.81		0.78			0.87		0.76
Attainment (Average)		0.81	0.65	0.64	0.62		0.68		0.69

Analysis & Continuous Improvement



- 📌 **Gap Analysis:** Identify areas where students struggle.
- 📌 **Pedagogical Improvements:** Adjust teaching methods, assignments, and assessments accordingly.
- 📌 **Action Plan:** Suggest remedial measures (extra sessions, additional practice problems, revised assessment methods).





SUMMARY

CLOs 4 to 6

Ist (Level I-3), IInd & IIIrd
(Level I-4), IVth(Level 5/6)

CLOs & PLOs Mapping

Ensure ≤ 2

ECs

Ensure Formative min 30% and
Summative max 70%



CLOs/ECs

Weightage Distribution 100%
CLO wise

QP Setting

RBT I- 4 (80%) and RBT 5-6(20%)

CLOs Attainment

<60 – Not attained(Risk)
 ≥ 60 to <80 – Satisfactory
 ≥ 80 – Excellent

PLOs Attainment

<60 – Not attained(Risk)
 ≥ 60 to <80 – Satisfactory
 ≥ 80 – Excellent

